INCH-POUND

A-A-59131B
May 14, 2014
SUPERSEDING
A-A-59131A
March 19, 2004

COMMERCIAL ITEM DESCRIPTION

DETECTOR METAL FLAW, EDDY CURRENT, IMPEDANCE PLANE

The General Services Administration has authorized the use of this Commercial Item Description, for all federal agencies.

1. SCOPE. This Commercial Item Description (CID) defines the minimum performance and technical requirements for a lightweight microprocessor-based eddy current flaw detection instrument suitable for Nondestructive Inspection (NDI) use in a military industrial maintenance environment, which may be at fixed military bases, deployed to austere forward operating locations, or naval vessels afloat. This CID provides for first article testing.

2. CLASSIFICATION. There are eight types and three classes of metal flaw detectors.

Type I	Basic Kit with case, charger, and accessories.
Type II	Intermediate Kit with case, charger, and accessories, with bolthole scanner.
Type III	Intermediate Kit with case, charger, and accessories, with bolthole scanner, aluminum bolthole standard, and probes.
Type IV	Intermediate Kit with case, charger, and accessories, with bolthole scanner, steel bolthole standard, and probes.
Type V	Intermediate Kit with case, charger, and accessories, with bolthole scanner, aluminum, steel, and titanium bolthole standards, and probes.
Type VI	Advanced Kit with case, charger, and accessories, with advanced bolthole scanner, aluminum, steel, and titanium bolthole standards, & probes, with automatic defect recognition.
Type VII	Advanced Kit with case, charger, and accessories, with bolthole scanner, aluminum, steel, and titanium bolthole standards, and probes, and surface area ("C") scanning capability.
Type VIII	Advanced Kit with case, charger, and accessories, with bolthole scanner, aluminum, steel, and titanium bolthole standards, and probes, and multi-channel array scanning capability.

Comments, suggestions, or questions on this document should be addressed to AFLCMC/ENRS, Bldg. 28, 2145 Monahan Way, Wright-Patterson AFB, OH 45433-7017 or emailed to AFLCMC/ENEZ.ENGINEERING.STANDARDS@US.AF.MIL. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 6635

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Class 1	Normal Industrial Maintenance Environment
Class 2	High Radio-Frequency (RF) Environment
Class 3	Chemical, Biological, Radiological, or Nuclear (CBRN) Environment

3. SALIENT CHARACTERISTICS.

3.1 General. The unit shall have a microprocessor-based, impedance plane display and have dual frequency capability. Type VIII shall have a minimum of 32 independent channels.

3.2 Foreign Object Damage. The unit shall be designed to prevent loss of components, which could generate foreign object damage to weapon systems or engines.

3.3 Size and Weight. The basic & intermediate unit size shall not exceed 432 cubic inches, with no dimension exceeding12 inches and the weight shall not exceed 7 pounds with batteries installed (not including scanners). Advanced units shall not exceed 1723 cubic inches with no dimension exceeding 18 inches and the weight shall not exceed 14 pounds with batteries installed (not including scanners).

3.4 Power Requirements. The unit shall operate when supplied with either 100-250 Volt Alternating Current – 50 or 60 Hertz (100-250 VAC – 50/60 Hz) automatically selected and connected directly, through means of a battery charger connected to 100-250VAC-50/60Hz, or batteries/battery pack(s) as described below.

3.4.1 Battery Charger. The unit shall be supplied with a separate battery charger(s) that is/are able to simultaneously charge two batteries/battery pack(s) and provide power to operate the instrument and charge installed batteries/battery pack.

3.4.2 Battery Duration. The cells in the rechargeable battery pack(s) shall not have a charge "memory" limitation that causes the unit to operate for less than the specified time, and provide at least 6 hours of continuous use without scanner or display backlighting, and at least 3 hours of continuous use with both scanner and backlighting at 68 degrees Fahrenheit. The unit must incorporate a visible means of indicating the charge condition of the battery pack(s) and shall provide a low battery warning in order to perform a post-calibration before the battery is exhausted. A minimum of four (4) complete sets of batteries/battery pack(s) shall be provided, in order to provide 24 hour a day operation.

3.4.3 Battery Charging. The batteries shall be easily rechargeable without tools and require no longer than 4 hours to charge from 50% to 100% capacity, nor more than 8 hours for a complete charge.

3.4.4 Battery Replacement. The units shall be capable of continuous 24 hour per day – 7 day per week operation, by swapping rechargeable battery packs by the user without tools and within 90 seconds while meeting the requirements of 3.3. If designed for use with other than common commercially available worldwide (e.g., AA or D cell) batteries, an auxiliary battery pack designed for AA or D cells shall be provided, which shall connect to the flaw detector. The auxiliary battery pack is not subject to the requirement of 3.3. If designed for proprietary battery packs, the government shall acquire the design specifications and a limited license for manufacture after contract expiration.

3.4.5 Battery Safety. The unit shall incorporate battery safety devices to prevent thermal runaway and cell rupture.

3.4.5.1 Air Transport. The extra sets of batteries/battery pack shall have a protective case to permit airline transport (see http://safetravel.dot.gov/ for additional information).

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3.4.5.2 Lithium Battery. Any lithium batteries used shall be in accordance with NAVSEA TM S9310-AQ-SAF-010. Detailed instructions for the approval process including testing requirements are contained in this technical manual. In some cases, the testing requirements can be waived if particular conditions are met, per NAVSEA TM S9310-AQ-SAF-010.

3.4.5.3 Waivers. The responsible government engineering office will coordinate with their respective safety organization, and provide any waiver recommendations to the Contracting Officer.

3.5 Controls. The unit function controls shall be menu driven, not to exceed two (2) sub-menu levels on basic and intermediate units. As a minimum, the unit shall accommodate control of the following functions.

3.5.1 Frequency. The operating frequency shall be selectable from 60 Hz to 6 MHz (minimum range), with a precision of two significant digits (i.e., 1 Hz steps from 60-100Hz; 10 Hz steps from 100Hz-1kHz; 100Hz steps from 1-10kHz; 1kHz steps from 10kHz-100kHz; 10kHz from 100kHz-1MHz; and 100kHz steps from 1-6MHz). The frequency synthesizer shall be accurate to within five (5) percent of the selected value. Sampling rate shall be a minimum of 500 Hz.

3.5.2 Gain. The unit shall be capable of at least 90 dB of amplification range for both the vertical (y) and horizontal (x) axes.

3.5.3 Probe Drive. The unit probe drive circuit shall allow the equipment operator to vary the signal driving the probe independent of gain and frequency adjustment. The unit shall operate wide-field ribbon probes without use of an external pre-amplifier.

3.5.4 Phase Rotation. The unit shall be able to rotate the eddy current impedance vector, as viewed on the display, through a full 360 degrees in one degree or smaller increments in either the clockwise or counter-clockwise directions.

3.5.5 Filters. The unit shall be equipped with high pass, low pass, and bandpass filters. The low pass filter shall be selectable from 10 Hz to 1500 Hz minimum range, and the high pass filter shall be selectable from 0 to 500 Hz minimum range.

3.5.6 Alarm Function. The unit shall incorporate an alarm (gate) system that will trigger an audible and/or visual alarm when the test signal enters or leaves a designated alarm zone. The alarm function shall be adjustable for size, position, and shape, and useable in both the impedance plane and sweep modes. The audio alarm frequency shall be lower than 2000 Hz and have an intensity of 50-70dB at three feet.

3.5.7 Programmable Test Procedures. The unit shall have the capability to program, store, and recall a minimum of 100 instrument test parameter settings. The instrument memory shall enable alphanumeric labeling of programs. Programmed instrument settings shall be adjustable after recall. The memory shall retain the settings for a minimum of 168 hours after electrical power is turned off and primary batteries are removed. The unit shall have a “load default” feature based on Technical Order (T.O.) 33B-1-2 settings, unless otherwise specified in the solicitation or post-award conference. These settings are available upon request.

3.5.8 Data Capture. The unit shall have the capability to freeze and store a minimum of 20 screen displays for subsequent review or download to computer or printer. The unit shall have the capability to capture a minimum of 60 seconds of impedance plane trace data (x/y versus time) for subsequent review or download to a computer. The unit shall have the capability to store a minimum of 20 trace data files.

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3.6 Instrument Display. The unit shall have a color display with a minimum area of 13 square inches having a minimum resolution of 76,800 total pixels. The display shall be easily seen both in bright light (i.e., flight line ~ 500 foot-candles) or low light (i.e., hanger ~ 10 foot-candles) conditions. The brightness level shall be adjustable. The refresh rate shall be a minimum of 60 Hertz. The display shall have a selectable graticule for both major and minor divisions.

3.7 Persistence. The display persistence shall be variable and adjustable from 0.1 to 10 seconds minimum range or until cleared by the operator.

3.8 Display Modes. All units shall display eddy current inspections signals in the impedance plane (flying dot) and the vertical amplitude versus time (sweep). Intermediate and advanced units shall have dynamic bolthole scanner display. Advanced units shall have Area "C" Scan Mode.

3.8.1 Impedance Plane Mode. The impedance plane mode shall display the eddy current signal as an impedance vector representing the vector sum of the test coil inductive reactance and resistance. The impedance plane mode shall be available for both manual and bolthole scanner modes. The display shall incorporate separate controls for horizontal and vertical positioning of the balance point anywhere on the screen.

3.8.2 Sweep Mode. The sweep mode shall display the vertical impedance plane component of the eddy current signal as a function of time. The unit shall be capable of synchronizing with an external signal.

3.8.3 Dynamic Bolthole Scanner Mode. The display shall have a dynamic mode that is compatible with the bolthole scanner defined in 3.11.1.1. The dynamic display shall automatically initiate a sweep trace at the start of each scanner probe rotation. Display options shall include showing only the current sweep (y versus t) while automatically erasing the previous sweep information, showing impedance plane and sweep, and a waterfall type display to show consecutive sweeps if connected to an indexing scanner. The full 360 degrees of rotation shall be displayed regardless of any nominal changes in rotation speed of the scanner probe.

3.8.4 Surface Area "C" Scan Mode: The advanced units shall display the vertical impedance plane component of the eddy current signal against an area scan using a variable color palate.

3.9 Conductivity and Coating Thickness Measurement. The unit shall be capable of measuring electrical conductivity and non-conductive coating thickness as specified below. The unit shall come equipped with appropriate probes and cables for conductivity and thickness measurement. The probe(s) shall be replaceable in the field without the need for depot or contractor intervention.

3.9.1 Electrical Conductivity. The unit shall be capable of measuring electrical conductivity of materials in the minimum range from 1 to 65% IACS (International Annealed Copper Standard). The unit shall be capable of using 60 ± 10 kHz and 500 ± 20 kHz for conductivity measurement. The measurement accuracy for the range from 20 to 65% IACS shall be within ± 0.5% IACS at 68 degrees Fahrenheit with liftoff up to 0.008 inches. The accuracy for the range from 1 to 65% IACS shall be within ± 1.0% IACS from 32 to 100 °F.

3.9.2 Non-conductive Coating Thickness. The unit shall be capable of measuring non-conductive coatings up to 0.015 inches thick. The measurement accuracy shall be within ± 0.001 inch of the reading.

3.10 External Connections. The unit shall accommodate the following external connections.

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3.10.1 Probe Interface. The unit shall have a 16-pin Fischer Lemo FGG.2B connector, unless otherwise specified or negotiated with the Contracting Officer. Typical pin out wiring is as follows:

Pin 1 – Signal ground
Pin 2 – Positive (+) Detector input
Pin 3 – Resistor drive for bridge probes
Pin 4 – Voltage drive for reflection probes and scanners
Pin 5 – ID chip
Pin 6 – Tachometer pulses
Pin 7 – Positive (+) Motor power for scanners
Pin 8 – Motor power return
Pin 9 – Index pulse
Pin 10 – Resistor drive for bridge probes
Pin 11 – Negative (-) detector input
Pin 12 – Ground to null
Pin 13 – Ground to erase
Pin 14 – Positive (+) 5V for alarm
Pin 15 – Positive (+) 5V for scanner
Pin 16 – Scanner direction

3.10.2 Probe Compatibility. The unit shall be electronically compatible with all currently available commercial off-the-shelf absolute, reflection, and differential-type probes.

3.10.3 Scanner Interface. The unit shall interface and be compatible with bolthole scanners. As a minimum, the eddy current unit shall provide the following: (1) scanner power; (2) sweep coordination (external triggering); (3) control of scanner probe revolutions per minute (RPM); (4) compatibility with remote null switching on the scanner; and (5) capability to change probes in the field. The scanner control capability shall be included with the unit and be ready for use. Types II through VIII shall include a high-speed bolthole scanner in the kit.

3.10.4 Analog Output. The unit shall have analog outputs to allow for horizontal and vertical component test signal monitoring.

3.10.5 Computer Interface. The unit shall include Microsoft Windows (7 or later version) compatible software and cables to enable communication with a personal computer. Software capabilities shall include: (1) software version updates; (2) uploading and downloading instrument program settings; and (3) data reporting utilities such as downloading screen captures or traces. While open architecture is preferred, if the unit does not store the data in a standard software format (e.g, .tif, .jpg, .txt, etc.), software shall be provided to convert the file format. The unit shall have USB connectivity.

3.10.6 Printer Output. The unit shall be capable of printing screen presentations and test procedures. This may be accomplished either directly to a USB printer or through a Microsoft Windows (7 or later version) compatible computer, with supplied software. The unit shall come with drivers for common commercially available printers.

3.10.7 Video Output. The unit shall have a Video Graphics Array (VGA) output for interfacing with a 15-pin DE-15 video projector connection to present the display in a classroom environment.

3.11 Accessories.

3.11.1 Mechanical Scanners.

3.11.1.1 Bolthole Scanner. Types II through VIII shall contain a high speed bolthole scanner. Type VI shall contain a bolthole scanner capable of automatic defect recognition. Bolthole scanners shall have a minimum speed range of 1000 to 2400 revolutions per minute and discrete speeds shall include 1200, 1500, 1800, and 2400 revolutions per minute, if not infinitely variable. Bolthole scanner size shall not exceed 6 x 3 x 2 inches, excluding the probe. Bolthole scanner shall have sufficient torque to rotate a

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0.750 inch diameter probe at 1500 revolutions per minute through the 0.875 inch thick aluminum plate assembly of the universal eddy current standard, NSN 6635-01-587-1418.

3.11.1.2 Area Scanner. Type VII shall contain an area scanner capable of scanning a 24 inch by 24 inch area, with a minimum resolution of 0.050 inches x 0.050 inches and manually operated vacuum attachments. Type VIII shall contain a linear scanner with an encoder capable of 0.050 inches minimum resolution.

3.11.2 Protective Carry Cases. The unit shall include soft-shell and hard-shell carrying cases. The soft shell case shall be ergonomically designed and protect the unit from damage during use while still allowing access to all external connections and have appropriate hand/neck straps to help prevent accidental loss. The hard-shell case shall be suitable for storing and shipping the unit with accessories. If the hard-shell case with unit and accessories weighs in excess of 25 pounds, the hard shell case shall include rollers and collapsible handle typically found on luggage. The hard-shell case size and total weight shall be limited to airline checked baggage restrictions of 50 pounds and dimensions of 62 total inches (length plus width plus height). The cases shall be designed to protect the unit from an 18 inch drop in soft case and a 36 inch drop in the hard case. The Type VII area scanner is allowed a separate case meeting the above requirements.

3.11.3 Cables. The unit shall have a cable to connect absolute “Microdot” type probes to the unit. The unit shall have a cable to connect Fischer Triax probes to the unit. Types II through VIII units shall have a cable to connect the bolthole scanner to the unit. The unit shall have a cable to connect a VGA projector to the unit. The unit shall have a cable to connect to a USB computer port. Cable lengths shall be between 6 and 10 feet.

3.11.4 Probes. All types shall contain a shielded surface probe kit, all with Fischer Triax connectors and internal balance coils: two (2) each 1/8 inch diameter stainless-steel shaft, straight pencil probes with 4 inches overall length with a frequency range of 50-500kHz; two (2) each 1/8 inch diameter stainless-steel shaft, 0.5 inch drop, 90 degree pencil probes with 6 inch overall length with a frequency range of 50-500kHz; two (2) each 1/8 inch diameter flexible copper shaft, straight pencil probes with 6 inches overall length with a frequency range of 50-500kHz; two (2) each 1/8 inch diameter stainless-steel shaft, straight pencil probes with 4 inches overall length with a frequency range of 1-3MHz; and two (2) each 1/8 inch diameter stainless-steel shaft, 0.5 inch drop, 90 degree pencil probes with 6 inches overall length with a frequency range of 1-3MHz. The unit shall have a conductivity/coating thickness probe to meet the requirements in 3.9. Conductivity probe diameter shall not exceed one-half (0.5) inch.

3.11.4.1 Probes – Types II through VIII. Types II through VIII shall contain bolthole probes which include a stainless-steel bolthole probe kit; with two (2) each probes for the following diameter holes: 0.125, 0.156, 0.187, 0.250, and 0.312 inches. The unit shall include a plastic bolthole probe kit for the scanner containing twenty (20) bolthole probes in sizes 5/32 inch to 3/4 inch in 1/32 inch increments, with its own separate ruggedized container. Working length shall be a minimum of 1.0 inches below 0.250 inches diameter and a minimum of 2.0 inches at or above 0.250 inches diameter. Bolthole probes shall have an operational frequency range suitable for the included reference standards for each Type.

3.11.4.2 Probes – Type VII. Type VII shall contain two each surface probes for the area scanner with a frequency range of 50-500 kHz.

3.11.4.3 Probes – Type VIII. Type VIII shall contain two each array probes for the linear scanner with a frequency range of 50-500 kHz for inspection of surface defects on aluminum, one each array probe for the linear scanner with a frequency range of 1-25 kHz for inspection of shallow sub-surface defects on

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aluminum, and one each array probe for the linear scanner with a frequency range of 0.2-10 kHz for inspection of deep sub-surface defects on aluminum.

3.11.5 Auxiliary Battery Pack. An auxiliary battery pack shall be provided which utilizes either "AA" or "D" cell batteries if not used as the primary battery.

3.11.6 Reference Standards. All units, except Type IV, shall include aluminum conductivity standards with conductivity of 30% ± 1% and 60% ± 1% IACS. Applicable bolthole and surface reference standard meeting the requirements of NSN 6635-01-587-1418, P/N NRK-3A or NSN 6635-01-092-5129, P/N 7947479-10, shall be provided according to Type. Alloys are 7075-series aluminum, 4340 steel, and 6AL-4V titanium, unless otherwise specified. Drawings are available upon request.

3.12 Environmental Conditions. The unit shall be operable in the following environmental conditions.

3.12.1 Explosive Atmospheres. The unit shall be tested and certified by a reputable laboratory for safe use, while operating on batteries, in an explosive atmosphere as defined by Class 1, Division 2, Group D, in the NFPA 70 Chapter 5, Article 500. The testing method shall be in accordance with MIL-STD-810, Method 511.5, Explosive Atmosphere, Procedure I, using a Fischer Triax eddy current manual surface probe and Fischer Lemo cable attached, in a typical operating condition.

3.12.2 Operating Conditions. The unit and accessories shall perform to the specification in this CID throughout temperatures from 15 to 120 °F at up to 90% relative humidity and from sea level to 5280 feet above sea level. For use below 50 °F, the unit shall start at an ambient temperature of 50 °F, and then shall operate on battery power for a minimum of 30 minutes at temperatures of 15 to 32 °F, and similarly for a minimum of 60 minutes at temperatures from 32 to 50 °F.

3.12.2.1 Normal Maintenance Environment – Class 1. The equipment shall operate in an industrial work environment.

3.12.2.2 High Radio-Frequency (RF) Environment – Class 2. The contractor shall test the unit's Electro-Magnetic Interference (EMI) capabilities in a high Radio-Frequency (RF) environment (e.g., above deck per MIL-STD-461) and provide a test report per DI-EMCS-80201 as part of first article testing to the Contracting Officer within 90 days of contract award.

3.12.2.3 Chemical, Biological, Radiological, or Nuclear (CBRN) Environment – Class 3. The equipment shall operate in a chemical, biological, radiological, or nuclear warfare environment with the operator in Mission Oriented Protective Posture (MOPP) Level 4 protective clothing. The CBRN test plan will be provided in the solicitation.

3.12.3 Storage Conditions. The unit and accessories shall perform to the specifications in this CID following a seven (7) day exposure to temperatures from 0 to 158 °F at up to 95% relative humidity with units in shipping cases.

3.13 Calibration. The unit shall be user calibrated without the need for manufacturer's periodic calibration. No factory recalibration decals shall be applied to the unit.

3.14 Software Revisions. Software revisions shall be user installable from a Microsoft Windows-based (7 or later versions) personal computer. All software revisions made during the life of the contract shall be included at no additional charge. A copy of revisions, and any necessary executable software, released

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during the life of the contract shall be provided to AFRL/RXS, Attention Webmaster, 4750 Staff Drive, Tinker AFB, OK 73145-3318, or as specified in the solicitation or contract, to ensure proper configuration management.

3.15 User's Documentation. One copy of the commercial operation and maintenance instructions, including calibration procedures and adjustments, as applicable, shall be shipped with each item. The documentation shall include information for battery safety and known environmental restrictions on battery disposal. The manual shall provide instructions for users to replace internal battery for clock and memory functions.

3.16 Provisioning. Two paper and two electronic copies of the operation and maintenance instructions (above), illustrated parts breakdown, diagnostics, and servicing instructions, shall be sent to the Contracting Officer for delivery to the Equipment Specialist and cognizant Engineer. The documentation shall be releasable for third-party repairs after expiration of the contract, subject to standard government non-disclosure agreements between original equipment manufacturers (OEMs) and third-party providers. Recommendations for long-term provisioning shall be sent to Contracting Officer for delivery to the Program Management Specialist, Inventory Management Specialist, Production Management Specialist, Equipment Specialist, and cognizant Engineer. For subsequent delivery orders, any changes to the instructions or recommended provisioning shall be sent to the Contracting Officer for similar distribution.

3.17 Warranty. A minimum one-year warranty, to include parts, labor, and two-way shipping, shall be included against manufacturing defects.

3.18 Options. A listing of all available accessory items, extended warranties, periodic recalibrations/recertifications, and standard repairs shall be provided to the Contracting Officer for government consideration (see DI-ILSS-80868). Pricing shall match the quantity breakdowns provided in the solicitation or contract.

3.19 Post-Award Conference. A post-award conference shall be held at the manufacturer's location after successful completion of first article testing to discuss potential customization of software, kitting changes, and certification of interface software for installation on government networked computers. A minimum of thirty (30) days advance notice shall be provided. Any changes affecting cost require approval by the Contracting Officer.

3.20 Sub-Contracting Plan. The proposal shall include a subcontracting plan that meets the requirements of 19.7 of the Federal Acquisition Regulation (FAR).

4. REGULATORY REQUIREMENTS. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with 23.403 of the Federal Acquisition Regulation (FAR).

4.1 Metric Products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pounds unit, provided they fall within specified tolerances using conversion tables contained in the latest revision of FED-STD-376, and all other requirements of this CID are met. If a product is manufactured to metric dimensions and those dimensions exceed tolerances specified in the inch-pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

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5. PRODUCT CONFORMANCE.

5.1 Product Conformance. The products provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance.

5.2 Market Acceptability (MA). All of the following market acceptance criteria are necessary to document the quality, reliability, and supportability of the product to be provided under this specification: (1) The company producing the item must have been producing eddy current flaw detectors for at least the past three years; (2) The company must have sold the basic commercial unit in the industrial market for a minimum of one year; and (3) The company must have sold at least fifty (50) units similar to the units being offered, with no major defects within the last year. A major defect is a defect, which affects the performance of the unit and requires an engineering change to the unit's original configuration in order for the unit to perform the requirements of this CID.

5.3 First Article Test. Requirements for first article testing, performed in accordance with 9.3 of the FAR, will be provided in the Solicitation.

6. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order.

7. NOTES.
(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

7.1 Part or Identification Number (PIN). The following PIN procedure is for government purposes and does not constitute a requirement for the contractor.

This example describes a part numbering system for CID A-A-59131B.

AA59139B 1 1 Example of reference part number:

AA59131B-1-1

Type:
- 1- Type I - Basic Kit with case, charger, and accessories
- 2- Type II - Intermediate Kit with case, charger, and accessories, with bolthole scanner
- 3- Type III - Intermediate Kit with case, charger, and accessories, with bolthole scanner, aluminum bolthole standard, and probes
- 4- Type IV - Intermediate Kit with case, charger, and accessories, with bolthole scanner, steel bolthole standard, and probes
- 5- Type V - Intermediate Kit with case, charger, and accessories, with bolthole scanner, aluminum, steel, and titanium bolthole standards, and probes
- 6- Type VI - Advanced Kit with case, charger, and accessories, with advanced bolthole scanner, aluminum, steel, and titanium bolthole standards, and probes, with automatic defect recognition
- 7- Type VII - Advanced Kit with case, charger, and accessories, with bolthole scanner, aluminum, steel, and titanium bolthole standards, and probes, and surface area ("C") scanning capability

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8- Type VIII - Advanced Kit with case, charger, and accessories, with bolthole scanner, aluminum, steel, and titanium bolthole standards, and probes, and multi-channel array scanning capability

Class: 1- Class 1 - Normal Industrial Maintenance Environment

2- Class 2 - High Radio-Frequency (RF) Environment

3- Class 3 - Chemical, Biological, Radiological, or Nuclear (CBRN) Environment

7.2 Source of Documents.

7.2.1 Copies of government documents. Copies of government documents are available online at http://quicksearch.dla.mil.

7.2.2 Copies of Technical Orders. Copies of Technical Orders are available from AFLCMC/EZGTP, at 405-736-7597 or DSN 336-7597.

7.2.3 Copies of the Federal Acquisition Regulations (FAR). Copies of the FAR documents are available online at http://acquisition.gov/far/index.html.

7.2.4 Copies of the National Fire Protection Association (NFPA). Copies of NFPA document are available online at www.nfpa.org or from NFPA, 1 Batterymarch Park, Quincy, MA 02169-7571.

7.2.5 Copies of Naval Sea Systems Command Technical Manual (NAVSEA TM). Copies of NAVSEA TM are available online at http://www.public.navy.mil/comnavsafecen/Documents/afloat/Surface/CS/Lithium%20Batteries%20Info/LithBatt_NAVSEA_TMS9310.pdf.

7.3 National Stock Numbers (NSNs). The following is a list of NSNs assigned that correspond to this CID. The list may not be indicative of all possible NSNs associated with the CID.

NSN	Type	Part Number	CAGE CODE
6635-01-159-0175RN	Type I	A-A-59131B-1-1	81348
TBD	Type II	A-A-59131B-2	81348
6635-00-463-1574RN	Type III	A-A-59131B-3-1	81348
TBD	Type IV	A-A-59131B-4	81348
TBD	Type V	A-A-59131B-5	81348
TBD	Type VI	A-A-59131B-6	81348
TBD	Type VII	A-A-59131B-7	81348
TBD	Type VIII	A-A-59131B-8	81348

7.4 Ordering data. The contract or order should specify the following:

a. Title, number, and date of this document.

b. Product conformance provisions.

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c. The solicitation will provide for any required independent-laboratory first article testing or government testing, as appropriate.

d. Packaging requirements.

7.5 Key Words.

Bolthole scanner
Dual frequency
Eddy current
Electrical conductivity
Electromagnetic testing
Impedance plane

7.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

MILITARY INTERESTS:

Custodians:
Army – GL
Navy - AS
Air Force - 11
DLA - GS

CIVIL AGENCY COORDINATING ACTIVITY:
GSA-FAS

Preparing activity:
Air Force - 11

(Project No. 6635-2014-001)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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